

Roll No.....

END SEMESTER EXAMINATION DECEMBER-2025

Name of the Course: B. Tech.

Semester: I

Name of the paper: Engineering Physics

Paper Code: TPH 101

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two sub-questions among a, b, & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each sub-question carries 10 marks.

Q1	(20 marks)	
(a)	In Newton's ring experiment derive the expression for diameter of bright and dark rings.	CO 1
(b)	In a wedge shape film deduce the expression for path difference between reflected light and write the condition of maxima and minima.	
(c)	Deduce the expression for resultant intensity in Fraunhofer diffraction due to N-slits.	
Q2	(20 marks)	
(a)	Write the differences between spontaneous emission and stimulated emission of radiation. Discuss the construction and working of Ruby laser also draw its energy level diagram.	CO 2
(b)	Discuss the resultant due superposition of two linearly polarized light vibrating in perpendicular plane and travelling through a thin calcite crystal with optic axis parallel to refracting faces.	
(c)	Define specific rotation. A sugar solution in a tube of length 15cm produces an optical rotation of 20 degrees. The solution is then diluted to one fourth of its previous concentration. Find the optical rotation produced by 30cm long tube containing diluted solution.	
Q3	(20 marks)	
(a)	Derive Lorentz transformation equations for space and time coordinates and show that these equations become Galilean equations at low speeds.	CO 3
(b)	Prove Einstein mass energy relation. Show that $E^2 = m_0^2 c^4 + p^2 c^2$	
(c)	Write Einstein postulate for special theory of relativity. What is the length of one meter stick moving parallel to its length when its mass is 3/2 times its rest mass?	
Q4	(20 marks)	
(a)	State de-Broglie hypothesis of matter waves. Calculate the ratio of de-Broglie wavelength of a hydrogen atom and helium atom at room temperature (27°C), when they move with their thermal velocities. Given Boltzmann's constant $k = 1.376 \times 10^{-23}$ J/K Planck's constant $h = 6.62 \times 10^{-34}$ J-s mass of hydrogen atom $m_H = 1.67 \times 10^{-27}$ kg mass of mass of helium atom $m_{He} = 4 \times 1.67 \times 10^{-27}$ kg	CO4
(b)	Find the normalized eigen wave function and energy of the particle bounded in a one dimensional infinite potential well or box; plot the graph of first three probabilities densities of particle within the potential well.	
(c)	(i) Explain Heisenberg's Uncertainty Principle and discuss its applications (any two). (ii) The speed of electron is measured to be 6.0×10^3 m/sec to an accuracy of 0.004%. Find the uncertainty in determining the position of this electron.	
Q5	(20 marks)	
(a)	Show that a superconducting material behaves like a diamagnetic material; also write differences between type I and type II superconductors.	CO5 & CO6
(b)	Derive all four Maxwell's equations write their physical significance.	
(c)	Name different techniques used for the fabrication of nanomaterial and give the details of any two methods; also write short notes on different types of nano structure.	

